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Signaling Quality: Dynamic Price-Advertising Model

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Abstract

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J Optim Theory Appl (2009) 143: 479496

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the signaling role of advertising by increasing spending to accelerate perceived quality increases. In cases when the value of the perceived quality goes up together with the increase in the perceived quality by more than the demand, in percentage terms, the rm should increase the price (use its signaling role). At steady-state, we nd that the level of optimal prot margin relative to price decreases with the elasticity of demand with respect to the brand price. However, higher elasticity of demand with respect to the rms perceived quality and/or a higher impact of price (advertising) lead/leads to a higher optimal prot margin (advertising spending) relative to price (revenue).

Keywords Marketing Perceived quality Price Advertising Signaling quality

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G.E. Fruchter

480 J Optim Theory Appl (2009) 143: 479496

1 Introduction

Consumers learn about the quality of a branded product through several mechanisms. In this paper, we assume that consumers learn about quality through advertising level and price signals.

Over the past several decades, researchers have focused their efforts on establishing, theoretically and empirically, the fact that price and advertising spending can be signals of product quality. They have adopted several different perspectives to explain the relationship between price and quality. Some studies focused on the relationship between objective product quality and price [13]. However, with the proliferation of the perspective that purchase decisions are not based on objective facts, but rather on subjective beliefs, the emphasis has shifted from objective product quality to quality as perceived by consumers, or what is called perceived quality (e.g., [48]). For a review on this subject, see [9].

Consumers use of price to determine their lasting product quality perceptions has been demonstrated in numerous empirical studies (e.g., [7, 1015]). Reference [16] assumes that, while the fact that consumer may purchase less of a brand when its price rises is immediately observed, an opposite effect, due to a perceived increase in quality when the price rises, is neither easily extractable nor observable. They provide an empirical approach to account for, in an unbiased way, the perceived quality effect on consumers choices.

Numerous experimental studies show that consumers infer higher quality from higher prices [17]. This inference is consistent with the ndings of several case studies. Diverse products such as fountain-pen ink, car wax [18], vodka, skis, and television sets [19] have been successfully introduced at high prices to connote high quality. A variety of empirical data is also available. Analyses of Consumer Reports data yield positive price-quality rank-order correlations for many products, and particularly for consumer durables [3, 20]. The market for en primeur wine (wine futures) in the Bordeaux region allows producers to sell wine that is still in barrels, and producers send quality signals to uninformed buyers. Reference [21] uses original data on Bordeaux wines to show that the pricing behavior of producers depends, to a large extent, on their reputation, and much less on short-term changes in quality, as measured by experts grades.

The inuence of advertising on the consumers perception of quality was also tested empirically by many researchers, among them, [22] who used data about domestic and foreign automobiles, and [23] who designed a laboratory experiment.

There is ample evidence in the marketing and economics literature that consumers view high price and advertising levels as indicators of high quality [2432]. However, apart from buying a familiar brand for its perceived quality, the consumer also sought the value achieved by being able to buy the brand at a cheaper price [33, 34]. Thus, price serves two distinct roles in consumers purchasing decisions. First, as a product attribute, price affects the perceived quality. Second, as a measure of sacrice, price serves as a benchmark for comparing utility gains with superior product quality.

Over time, due to consumer fatigue or forgetfulness, consumers perceptions of product quality may decline. The questions are: How should a rm, on the one hand, design its pricing and advertising policy in order to counteract such

deterioration, and, on the other hand, maximize its cash ow over time. The aforementioned

J Optim Theory Appl (2009) 143: 479496 481

and far-reaching evidence about diverse price and advertising effects on consumers, together with efcient empirical approaches to measuring them, provide a partial answer. However, the phenomenon has not yet been modeled sufciently. Further research is therefore required to develop efcient pricing and advertising strategies that take into account consumers changing perceptions about quality.

In the current paper, using optimal control we provide normative decision rules for a rms pricing and advertising strategies so as to maximize prots over time, taking into account the fact that demand is affected by price and perceived quality. In addition, we consider the idea that perceived quality is continuously affected by price and advertising level signals. Thus, this paper extends the existing quality-signaling literature by investigating the roles of price and advertising levels as quality indicators in a dynamic framework. Thus, the author anticipates that the paper will make several contributions to the marketing literature.

First, we formally integrate long-run price and advertising effects on sales through perceived quality together with the usual short-run price effects on sales. Second, assuming perceived quality to be a form of goodwill, we modify the well-known [35] dynamic model to include price effects. By formulating an optimal control problem, we were able to develop price and advertising strategies as a function of perceived quality over time, i.e., to nd feedback solutions.

The feedback solution has a very interesting pattern. When prices play a dual role by initially signaling high quality, they are set at a high level, and later increasing demand when they are reduced, and when advertising is used only as a quality indicator, the rm should increase advertising spending to accelerate perceived increases in quality. On the other hand, for a multiplicative separable demand, as long as the value of the perceived quality rises together with the increase in the perceived quality by more than the demand, in percentage terms, the rm should increase the price.

At steady-state, we nd that the level of the optimal prot margin relative to the price decreases with the elasticity of demand with respect to the brands price. However, higher elasticity of demand with respect to the rms perceived

quality, and/or higher impact of price (advertising level), leads to a higher optimal prot margin (advertising spending) relative to price (revenue). Thus, taking into account both roles that price plays, the effect of price elasticity is reduced when the additional attribute created by the long-run effects of price is considered, increasing product differentiation. Also, for products where pricing (advertising level) is more (less) effective in terms of signaling quality, the optimal result is achieved when higher price levels are implemented and advertising expenses are lowered, and vice versa.

The remainder of this paper is organized as follows. In Sect. 2, we present the model and the notation. In Sect. 3, we present the optimal policy. In Sect. 4, we present our concluding remarks and future directions.

2 Model Formulation and Notation

We consider a monopolist rm that needs to decide on pricing and advertising strategies for a certain branded product, over time. The rm uses advertisements to present product information that would primarily elicit benecial associations (advertising-based product quality information or goodwill advertising, which leads to increased

valuation of products). In other words, advertising expenditure is used only as a signalling device, and thus it is purely a dissipative expense. Consumers can infer the products quality, or more exactly its perceived quality, by the advertising outlay. However, since consumers view the price of a brand product both as a monetary constraint and as a signal of its quality (informational or signal role of price), the rm should explore the dual role of price, both as a product attribute and as a monetary constraint. As a monetary constraint, price affects sales in the short-run. As a product attribute, price affects the perceived quality of the brand; and perceived quality affects sales in the long-run.

The existence of carryover price and advertising effects leads to an accumulation of the brands perceived quality over time. This necessitates modelling the relationship among price, advertising level, and perceived quality in a dynamic framework. We present this relationship next.

2.1 Dynamics of Perceived Quality

Let $x = x(t)$ be the perceived quality offered by the particular brand at time t. This

perceived quality summarizes the previously perceived quality of similar products sold under the same brand name, as well as all past signaling effects of such products prices and advertisements. Thus, it can be thought of as the accumulated goodwill of the brand at time t. We assume that in the absence of advertising and price effects, the perceived quality depreciates as a result of consumer fatigue regarding the brands goods.

We assume that u = u(t) represents the impact of advertising spending on per

ceived quality, at time t. Let p = p(t) be the price of the brand at time t.

In order to model these assumptions and relationships in a dynamic framework, we modify the well-known [35] model to the following dynamic equation:

x(t) = kp(t) + u(t) x(t), x(0) = x0. (1)

The left-hand side of (1) represents the change in perceived quality of the particular brand. The rst term on the right-hand side represents the direct impact of current price levels on the brands perceived quality. The second term reects the direct impact of advertising efforts on perceived quality, represented by u(t). The third term reects the rate at which the perceived quality of the brand deteriorates, represented by x(t) where we assumed, as in [35], that the rate of deterioration is proportional to the level of perceived quality. Thus, the constants k, , represent the direct price and advertising effects on the brands perceived quality, respectively, and the rate of perceived quality deterioration without those effects. For the given k, , , (1) states that a change in perceived quality will be positive (negative) if the impact of the current brands price and advertising on its perceived quality, kp(t) + u(t) is higher

(lower) than the impact of consume fatigue on its current perceived quality, x(t). In other words, kp(t) + u(t) represents the current price and advertising contribution

to the brands perceived quality, as a result of the price and advertising decisions at time t, while x(t) represents the current perceived quality loss as a result of consumer fatigue regarding the brand goods. The particular case where k = 0 and

= 1 reduces to the [35] model. If pss, uss, and xss represent the price, advertising

J Optim Theory Appl (2009) 143: 479496 483

impact, and perceived quality at steady-state, then the value kpss + uss will be ex

actly equal to the value of xss. Thus, at steady-state, equilibrium will exist between the value added and the value lost.

Note that the dynamic setting in (1) is designed to capture the effects of pricing and advertising level as a signaling device of perceived product quality in rst purchases (durable goods). In contrast to durable goods, consumers learn about the true quality of non durable goods by consuming them (see [36, 37]). Thus, in repeat purchases, the effects of pricing and advertising levels as a signaling device are diminished by the learning process.

2.2 Firms Problem

In an attempt to formulate the problem for the rm, we assume that the rate of sales S = S(t) is governed by the long-run component (perceived quality) x and the short-

run component (price) p as

S = S(p, x). (2)

It is important to note that since advertising has a dissipative role only the sales do not depend directly on the advertising level.

Hence, price in our model has a dual role: (a) an informational role about the perceived quality, i.e., its contribution to sales is through the perceived quality and(b) a constraint role, i.e., its contribution to sales comes directly through the demand curve. Advertising has a single role: a signaling role about perceived quality, i.e., its contribution to sales is through the perceived quality. We assume that the rate of sales decreases with the rms price (following the classical demand function), but increases with the brands perceived quality. This leads to the following conditions on the rst partial derivatives of S:

S/p < 0 and S/x > 0. (3)

Another assumption is that the production cost of a unit sale of the brand remains constant. Let us denote this constant by c. Thus, the gross prot equation at time t will be

R(p, u, x) = (p c)S(p, x). (4)

As is commonly stated in marketing literature, we assume that the impact of advertising expenses is to decrease protability. With this in mind and for simplicitys sake, we model the rms advertising expenses as .5u2.

Assuming that the monopolist maximizes the present value of net prot streams discounted at a xed discount rate r with respect to price and advertising levels over an innite horizon, his/her problem becomes

max

p,u

= [integraldisplay]

0

[(p(t) c)S(p(t), x(t)) .5u2(t)]ertdt[bracketrightbigg],

s.t. x(t) = kp(t) + u(t) x(t), x(0) = x0. (5)

In the control theory framework, u and p are control variables and x is a state variable.

484 J Optim Theory Appl (2009) 143: 479496

3 Optimal Policy

To solve the problem in (5), we employ dynamic optimization techniques (as in, [38]). This is accomplished by constructing the Hamiltonian of the rms problem, and then using it to obtain the necessary conditions for the optimal solution. We dene the current-value Hamiltonian H as

H = (p c)S(p, x) .5u2 + [kp + u x]. (6)

In (6), we introduced the new variable , which is the current adjoint variable, representing the shadow price associated with a unit change in the value of the perceived quality x, at time t. In other words, is the net benet or loss to the rm generated by improving perceived quality by one more unit at time t. The adjoint variable (co-state) is assumed to have a continuous rst derivative. The Hamiltonian in (6) can be interpreted as the instantaneous prot rate, which includes the value x of the change

in perceived quality x, created by the effect of price and advertising levels.

Denition 3.1 We term

= (x/S)S/x (7) the elasticity of demand with respect to the rms perceived quality.

Denition 3.2 We term

= (p/S)S/p (8) the elasticity of demand with respect to the rms brand price.

As is commonly stated in the pricing literature (e.g., [39]), it is assumed that > 1. Considering (6)(8), the necessary rst-order optimality conditions are stated in Theorem 3.1.

Theorem 3.1 Consider the prot maximization problem (5). At any time t, t > 0, the optimal pricing and advertising strategy (p, u) satises the following necessary conditions:

p =

where

1[parenleftbigg]

c +

k

(Sp)[parenrightbigg]

,

u = , (9)

x = kp + u x, x(0) = x0,

= (r +) (p c)

S(p, x)

x , lim

t

ert(t) = 0. (10)

J Optim Theory Appl (2009) 143: 479496 485

Proof See Appendix.

The second row in (10) corresponds to the equilibrium relation for investment in perceived quality (compare with capital goods as discussed in [40]). In our context, the equilibrium is in relation to the quality implied by the brands price and the advertising level spending. It states that the marginal opportunity cost (r +)

of investment in perceived quality should equal the marginal prot (p c)S(p,x)x

from increased perceived quality and the value gain

. In other words, the differential equation in the adjoint variable tells us what the value of increasing the perceived quality at each point of time is.

If one solves the TPBVP (two-point boundary value problem) equation (10) and substitutes the solutions x(t), (t) into (9), one nds an optimal strategy (p, u) as a function of time. Thus, in this case, the solution would tell us what the price and advertising levels would be at each point in time.

Myopic Firm vs. Foresighted Firm The optimal policy in (9) includes the case where the monopolist maximizes the integrand in (5) and disregards the dynamic equation (1). We say in this case that the monopolist behaves myopically. In other words, myopically, the monopolist disregards the effect of the current price and advertising levels on the perceived quality in the future. That is to say, the rm disregards the carryover price and advertising level effects. In such a case, the monopolist sets price and advertising levels at p =

1 c, and u = 0. Thus, myopic behavior corresponds

to = 0. Alternatively, the optimal policy in (9) modies the myopic decision rules

to account for long-run price and advertising effects, k > 0 and > 0 through . For example, if there is a benet to be gained from improving the rms perceived quality (> 0), the policy must be modied to account for this, causing prices and advertising levels to be higher for products where price and advertising signals (inuence) quality (thus k > 0 and > 0). Therefore, the sign of the shadow price determines whether the price and advertising levels of a foresighted rm will be higher or lower than the levels of a myopic rm

policy facing the same conditions. If > 0, then the strategic price and advertising levels will be higher. In the next proposition, we study the sign of .

Proposition 3.1

(i) (t) > 0, t.

(ii) The optimal price and advertising level is above the myopic price and the advertising level (price and advertising that corresponds to = 0).

Proof See Appendix.

Proposition 3.1 states that the brands optimal price and advertising levels, which account for the effect of current pricing and advertising strategy on the future perceived quality, as well as the long-run effects of price and advertising levels, will be higher than the myopic price and advertising levels. This result is consistent with a brand differentiation policy that facilitates higher prices and higher advertising spending to sustain the differentiation.

486 J Optim Theory Appl (2009) 143: 479496

Special Cases The optimal price and advertising levels in (9) include the following special cases.(i) Only advertising signaling effects, i.e., k = 0. In this case, the dynamic equa

tion (1), after normalizing to 1, reduces to the classical Nerlove-Arrow equation, corresponding to the case when goodwill is created only by advertising. In this case, (9) is reduced to

p =

1

c and u = .

Therefore, in this situation the rm behaves myopically with respect to price, but recognizes that investing in advertising level increases its brand value and, in turn, its prots.(ii) No advertising signaling effects, i.e., = 0. In this case, (9) is reduced to

p =

1[parenleftbigg]

c +

k

(Sp)[parenrightbigg]

and u = 0.

This situation is suitable for a foresighted rm that has a well-established brand for which the advertising level does not affect perceived quality.

3.1 Steady-State Solutions

In this section, we attempt to understand better the long-term behavior, that is, the behavior of solutions such as t , at steady-state. Such behavior is especially

important for well-established brands.

Theorem 3.2 Consider the prot maximization problem (5) and Denitions 3.1 and 3.2 at steady state. Suppose that the pair (pss, uss) satises the set of two algebraic equations

(pss c) pss =

1

(ss

ss

(1+ r)

1 1+ u

ss kpss)

and (uss)2(pss c)Sss =

ss

(1 + r)

1

(1 + kp

ss

uss)

,

(11)

in pss and uss. Then, the pair (pss, uss) forms a steady-state optimal strategy of the prot maximization problem (5).

Proof See Appendix.

Theorem 3.2 gives us a decision rule as follows:

(i) The level of prot margin relative to price in the long-run depends on the relative effect of the elasticity of demand with respect to the rms brand price ss on the

elasticity of demand with respect to the rms perceived quality (weighted by a

factor scaling long-run advertising and price), ss

(1+

r)

1 1+ u

ss kpss .

(ii) The level of advertising expenses relative to revenues in the long-run depends on the elasticity of demand with respect to the rms perceived quality (weighted by a factor scaling long-run price and advertising), ss

(1+

r)

1 (1+ kp

ss uss) .

J Optim Theory Appl (2009) 143: 479496 487

Considering the RHS of (11) and taking the partial derivatives with the relevant parameters, we obtain the following strategic implications at steady state:

(a) Higher elasticity of demand with respect to price, ss, permits a lower optimal prot margin relative to price.

(b) On the other hand, higher elasticity of demand with respect to the rms perceived quality ss and/or a higher (lower) impact of price (advertising) and/or a higher (lower) fatigue rate (discount rate) lead/leads to a higher optimal prot margin, relative to price.

(c) Higher elasticity of demand with respect to the rms perceived quality ss

and/or a higher (lower) impact of advertising (price) and/or a higher (lower) fatigue rate (discount rate) lead/leads to higher optimal advertising spending, relative to revenue.

Thus, taking into account both the two price roles and the signaling role of advertising, the effect of price elasticity is reduced by taking into account the additional attribute created by the long-run effects of price and advertising. The substitution effect is diminished and the prot margin goes up relative to price as the elasticity of demand with respect to the rms perceived quality increases, and/or the impact of price (advertising) in signaling quality increases (decreases). On the other hand, increased elasticity of demand with respect to the rms perceived quality increases advertising expenses (devoted to the signaling role) relative to revenue. These expenses should rise if the advertising level (price) has a higher (lower) signaling effect on quality. Thus, for products where pricing (advertising level) is more (less) effective for signaling quality, it is optimal to implement higher levels of price and to lower advertising expenses, and vice versa.

In other words, the consideration of the dual role of price and the signaling role of advertising creates a mechanism of product differentiation. This is consistent with the industrial organization literature on product differentiation (see, [4146]). On the other hand, increased elasticity of demand with respect to the rms perceived quality increases advertising expenses (devoted to the signaling role) relative to revenue. These expenses should rise if the advertising level (price) has a higher (lower) signaling effect on quality. Thus, for products where pricing (advertising level) is more (less) effective for signaling quality, it is optimal to implement higher levels of price and to lower advertising expenses, and vice versa.

3.2 Feedback Solutions

A solution that is specied as depending on the perceived quality x rather than directly on t is referred to as a feedback or state dependent solution. This is the type of solution we seek in this section. Thus, we solve for p and u as functions of the state variable x. Note that, since feedback solutions depend on the state variable, they provide a very practical control rule that allows the manager to adjust the price and advertising expenditure according to the perceived quality.

Sufcient Conditions In addition to the necessary conditions, we must also verify second-order conditions to ensure that we are indeed maximizing prots. Such a sufcient condition for local optimum (see [47]) is that the Hessian matrix of H , denoted

488 J Optim Theory Appl (2009) 143: 479496

by , is negative denite, that is,

H = [bracketleftBigg]

2Sp + (p c)Spp 0 0 1

[bracketrightBigg] < 0, (12)

for all (p, u) near (p, u). In (12), the subscript of S denotes a partial derivative. This requires that the principal minors of matrix alternate in signs, beginning with

the negative.

The following theorem presents a general form solution for the feedback optimal solution, which can then be employed to compute a closed form optimal strategy by examining the sufcient conditions for equilibrium.

Theorem 3.3 Consider the optimal control problem stated in (5) and assume that the sufcient condition holds in some neighborhood of (p, u). Assume also that the costate variable is a function of the state variable x, thus = (x). Then, the

necessary conditions dene a unique local time-invariant feedback optimal strategy for price and advertising of the form

p = p(x, (x)) and u = (x), (13) where (x) satises the following backward differential equations:

(x)(kp + 2 (x) x) = (r +) (x) (p c)

S(p, x)

x , (xss) = ss,(14)

and xss, ss are the steady state values of the state and costate.

Proof See Appendix.

In (14), xss and ss are solutions of x = 0 and

= 0.

The solution based on Theorem 3.3 is illustrated with an example in Sect. 3.3. Note that the time trajectories of our specic feedback strategy are p(x(t)) and u(x(t)), when x(t) is as in (10).

Characterization of the Feedback Solution Consider the feedback solution in (13). In the following proposition, we nd under which conditions the optimal price p

decreases or increases with the increase of perceived quality x.

Proposition 3.2 For the multiplicative separable demand,1

S(p, x) = f (x)g(p), (15) the optimal price is decreasing, when the perceived quality is increasing, if and only if

S/x

S >

(x)

(x) . (16)

1See for example [39].

J Optim Theory Appl (2009) 143: 479496 489

Proof See Appendix.

Since (x)

(x)

=

x , the right hand side of (16) can be interpreted as the change in

the value of the perceived quality, or the potential contribution to the rms prots, of improving the perceived quality by one unit, in percentage terms. Thus, the inequality (16) expresses that the demand rises by more than the value of the perceived quality in percentage terms. Accordingly, the implication of Proposition 3.2 follows. If demand increases with the increase of perceived value by more than the value of the perceived quality, in percentage terms, then the rm should use the role of price as a monetary constraint and improve prots by increasing demand by reducing price. On the other hand, suppose that the value of the perceived quality increases with the increase of the perceived quality by more than the demand, in percentage terms. In this case, the rm should use the other role of price as signaling quality and improve prots by increasing the prot margin through price increase.

Considering again the feedback solutions in (13) and assuming that the perceived quality x = x(t) increases over the time for all t, and the optimal price p decreases

with the increase of the perceived quality x, we obtain the following result.

Proposition 3.3 Consider the feedback solutions in Theorem 3.3 and assume that the perceived quality increases over the time for all t and the optimal price p decreases with the increase of the perceived quality. Then:

(i) (x) > 0 and

(t) > 0, t.

(ii) The optimal advertising u increases with the perceived quality.

Proof See Appendix.

This proposition states that when both price and advertising levels are used to signal quality and when prices are reduced with the increase in the perceived quality, then the rm should increase advertising spending. Thus, consumers can infer the products quality by the advertising expenditure on it. Moreover, the value of increasing the perceived quality also increases.

By using Propositions 3.2 and 3.3 we can characterize how a rm can optimally engage its price and advertising levels in signaling quality over time in the following way. For a multiplicative separable demand, as long as demand increases with the increase in perceived quality by more than the value of the perceived quality in percentage terms, then, the optimal policy should be to set a high price initially that signals high quality. Later, the rm should use the role of price as a monetary constraint to increase demand by reducing price and increase advertising spending to increase perceived quality.

To illustrate our solutions and acquire more insights, we next consider an illustrative example.

3.3 Illustrative Example

We now provide a numerical example to illustrate how we use Theorem 3.3 to obtain the optimal feedback solution and how it follows Propositions 3.2 and 3.3. For the

490 J Optim Theory Appl (2009) 143: 479496

purpose of illustration, we consider a multiplicative separable demand function as in Proposition 3.2 with the following specication,

S = S(p, x) = (a1x + b1)(a bp). (17)

Applying Theorem 3.3, we solve for the optimal policy and obtain

p(x) =

a + bc

2b +

k (x) 2b(a1x + b1)

and u(x) = (x), (18)

where (x) satises

(+ r) (x)

a1[(a bc)2(a1x + b1)2 k2 2(x)]

4b(a1x + b1)2

[braceleftbigg]2 (x) x +

k[(a + bc)(a1x + b1) + k (x)]

2b(a1x + b1) [bracerightbigg]

(x) = 0,

(xss) = ss. (19)

To nd (xss, ss), we solve (10), considering (17), for x = 0 and

= 0. Thus,

ss2 xss + k[parenleftbigg]

kss

2b(a1xss + b1) +

a + bc

2b

[parenrightbigg] = 0,

ss(+ r) + a1[parenleftbigg]

kss

2(a1xss + b1) +

a + bc

2 a[parenrightbigg]

[parenleftbigg]

kss

2b(a1xss + b1) +

a + bc

2b c[parenrightbigg] = 0. (20)

We chose the following set of parameters:

k = 1.1, = 0.009, = 0.1 (dynamic equations parameters),a = 1, b = 0.5, a1 = 5, b1 = 0.1 (demand parameters),c = 0.5, r = 0.1 (cost and discount rate parameters). (21)

Now, using a symbolic software, e.g., Mathematica [48], one can solve both (20) and (19), thus obtaining (18). We display the results in Fig. 1.

Note that we veried that for the set of parameters in (21), condition (16) is satised. As can be seen in Fig. 1, the feedback solution has a very interesting pattern. As the perceived quality increases, the optimal policy should be to set initially a price that signals high quality. Then, over time, the product price should decrease. On the other hand, the optimal advertising spending should be positively correlated to perceived quality; as the perceived quality increases, the rm should spend more. The intuition behind this characterization is as follows: given the signaling role of price, a higher price at the beginning will help the rm to support its reputation for quality. Thus, even if the price will later be reduced to affect demand (using its dual role), the perceived quality will increase; this increase is accelerated by the increase

J Optim Theory Appl (2009) 143: 479496 491

Fig. 1 Price and advertising as a function of perceived quality

in advertising. Since advertising is used in our model only as a signaling device and consumers can infer the products quality by the advertising expenditure on it, as the quality increases, rms should spend more.

4 Conclusions and Future Directions

In the current paper, we have proposed an analytical model that presents a formal examination of how a rm should use optimal pricing and advertising strategies over time as quality signaling devices.

The optimal policy provides a very practical control rule, which allows the manager to adjust the price and advertising expenditure according to the perceived quality. Furthermore, the analysis also suggests how to optimize the tradeoff between price and advertising as signaling quality. More specically, when perceived quality increases and price acts as a monetary constraint to increase demand, the rm should increase advertising spending to accelerate increases in perceived quality. When the value of the perceived quality rises together with the increase in perceived quality by more than the demand, the rm should increase price. The implications of these results are

the following. When the rm takes advantage of the monetary role of price and the signaling role of advertising, it improves prots by increasing the demand through both price and perceived quality (through advertising spending); and when the rm takes advantage of the signaling role of price it should improve prots by increasing the prot margin.

At steady-state, we demonstrate that taking the dual role of price and the signaling role of advertising into consideration creates a mechanism of product differentiation.

Future research could extend our analysis in several directions. First, in our model the advertising spending is just a signaling device; it is a purely dissipative expense. Future research would benet by exploring a dual role for advertising, to raise awareness about the product and to signal its quality. In this case, the demand will be an explicit function of price, perceived quality, and advertising. It would be interesting to explore whether the rms optimal advertising policy would change in this case.

492 J Optim Theory Appl (2009) 143: 479496

Second, in our model cost is considered a xed variable. A possible future direction could relate to situations where cost depends on quality. Such an extension is especially important for products whose costs increase with their true quality. These directions could also be further extended by considering competition and repeat purchases, wherein consumers learn about products true quality through consumption.

Appendix

Proof of Theorem 3.1 Considering (6), the necessary rst-order optimality conditions are H/p = 0 and H/u = 0, namely,

S + (p c)S/p + k = 0, (22)

u + = 0, (23)

plus the adjoint equation with the terminal condition,

= r

H

x = (r +) (p c)

S

x , lim

t

ert(t) = 0. (24)

Equations (22)(23) can be rewritten in terms of elasticity of demand (Denition 3.2) to obtain (9), where x is as in (5) and is as in (24).

Proof of Proposition 3.1 Considering (10), at steady state we obtain

ss =

(pss c)S(p

ss ,xss) x

+ r

.

Considering (3), we obtain

ss > 0.

Now, we argue that

> 0, t 0. (25)

Assume by contradiction that (25) is not true. Then, there is some t0 < such that

(t0) < 0.

Moreover, since the costate is assumed to be continuous, there is > t1 > t0 such

that

(t1) = 0 (and (t1) < 0). (26)

Considering (10) and (3), at t1 we have

|t=t1 < 0. (27)

J Optim Theory Appl (2009) 143: 479496 493

However, since t1 is in the neighborhood of t1, such that t1 > t1, (27) contradicts our assumption that 0 = (t1) > (t1). Thus, (25) is true. Considering (9), it is now

easy to conclude insistently with our proposition.

Proof of Theorem 3.2 Consider the Denitions 3.1 and 3.2 at steady state, then

ss = (xss/S(pss, xss))S(pss, xss)/x (28)

and

ss = (pss/S(pss, xss))S(pss, xss)/p, (29) where xss satises

kpss + uss xss = 0. (30)

Substituting (28)(30) and (24), at steady state, into the necessary conditions (22)

(23), we obtain

1 +

(pss c) pss

[parenleftbigg]ss +

ss

(1 + r)

kpss

kpss + uss [parenrightbigg] =

0

and

uss (pss c)Sss

ss

(1 + r)

(kpss + uss) =

0.

Proof of Theorem 3.3 Consider the necessary conditions (22)(23) and the sufcient condition (11). Next, we apply the implicit function theorem (see [49, p. 374]) to the system (22)(23) to arrive at the unique pair (p, u) in terms of x and ,

p = p(x,). (31)

Let

= (x). (32)

In particular, (32) results in

(x) x =

. (33)

Now, the theorem follows immediately from (10) after substituting (31)(33).

Proof of Proposition 3.2 Taking the derivative of (22) with respect to x, considering (32) and (15), and denoting the partial derivatives of S via subscripts, we obtain

px = k[bracketleftbigg]

(Sp) +

Spx

S2p

.

Thus,

sign[px] = sign[bracketleftbigg]

+

Spx

Sp

,

494 J Optim Theory Appl (2009) 143: 479496

where sign[] is 1 for a negative argument and +1 elsewhere. Considering

again (15), we have

Spx

Sp =

Therefore,

This completes the proof of this proposition.

Proof of Proposition 3.3 Substituting the solutions p and u from (13) into (1), and since x(t) increases for all t, we obtain

x = kp(x, (x)) + 2 (x) x > 0.

We want to show that if p decreases with the increase of x then u should increase.

Assume in contrast that u decreases, but since u = (x), this means that (x)

decreases when x increases. Since the last term becomes more negative and since p decreases, the rst term becomes smaller; thus if (x) also eventually decreases, the inequality will become zero and then negative, thus x < 0. However, this contradicts

our assumption that x = x(t) increases over time (x > 0) for all t. Thus (x) should

increase with x, and so u. This also implies (x) > 0 and (x) x =

> 0. This

completes the proof of this proposition.

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Sx

S .

sign[px] = sign[bracketleftbigg]

Sx

S

.

J Optim Theory Appl (2009) 143: 479496 495

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